



Spatial components of molecular tissue biology

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Methods for profiling RNA and protein expression in a spatially resolved manner are rapidly evolving, making it possible to comprehensively characterize cells and tissues in health and disease. To maximize the biological insights obtained using these techniques, it is critical to both clearly articulate the key biological questions in spatial analysis of tissues and develop the requisite computational tools to address them. Developers of analytical tools need to decide on the intrinsic molecular features of each cell that need to be considered, and how cell shape and morphological features are incorporated into the analysis. Also, optimal ways to compare different tissue samples at various length scales are still being sought. Grouping these biological problems and related computational algorithms into classes across length scales, thus characterizing common issues that need to be addressed, will facilitate further progress in spatial transcriptomics and proteomics.

Function in multicellular organisms depends on the balance of interactions between cells in tissues, which are complex, structured and dynamic cellular ecosystems. In health, tissues maintain homeostasis through the joint action of multiple cells, with a dynamic division of labor between different parenchymal and accessory cell types¹. In disease, malfunction often spans cells of multiple types and is accompanied by changes in the composition, structure and organization of tissues. Deciphering the relation between structure and function in tissues is the cornerstone of tissue biology and pathology, because the way in which cells and molecules are organized in histological patterns at different scales often reflects their joint functionality.

In recent years, there has been a dramatic growth in methods for spatial molecular profiling, which vary in resolution, scale and molecular multiplexing. Methods capture diverse information across different length scales: from single-molecule resolution in techniques such as MERFISH² or SeqFISH³ to averages across cells spanning dozens of micrometers in spot-based protocols such as spatial transcriptomics⁴. Methods also vary in the number of molecular features acquired: from tens in fluorescence in situ hybridization (FISH), cyclic immunofluorescence (CyCIF)⁵ and imaging mass cytometry (IMC)⁶ to hundreds or thousands in specialized probe-based spatial transcriptomics methods (MERFISH² or SeqFISH³) or imaging mass spectrometry⁷, and tens of thousands in spot-based spatial transcriptomics, such as Slide-Seq^{8,9}, Visium¹⁰, DBiT-seq¹¹ and high-definition spatial transcriptomics (HDST)¹². Such examples highlight a key feature of current spatial technologies: they are diverse in terms of resolution, throughput and multiplexing and should, accordingly, be used to address different classes of questions.

Computational methods are key to extracting patterns from such data and will be especially powerful if they are designed to take into account the specific biological questions at hand as well as the distinct features and limitations of different measurement methods. Here, we review the computational methods for spatial molecular analysis organized by the biological questions they address and the spatial methods capable of measuring relevant parameters. We will focus specifically on the challenges that different length scales

pose for experimental methods, and highlight the types of analysis method that can be deployed in such studies. We define ‘length scales’ as the spatial context in which a biological process occurs: short-range length scales include direct cell–cell interactions, whereas long-range length scales include global gradients, such as in oxygen or metabolites (Fig. 1). In addition, we emphasize conceptual overlaps with and distinctions from computational methods currently used for analyzing single-cell, dissociation-based methods to show how studies based on single-cell profiling approaches can be complemented by spatial approaches, and vice versa. We hope this conceptual and methodological roadmap will help drive the development of new computational methods for key biological questions in tissue biology, provide guidance to biologists seeking to apply methods and help in sharpening concepts in cell and tissue biology.

Modeling variation at length scales for cell and tissue biology

A long-standing goal in biology is to understand how tissue organization (‘structure’) relates to tissue physiology (‘function’). In a cell-centric model, tissue organization can be described by the different properties (or variables) that distinguish cells from each other (Fig. 1a). Some of these components can be seen as dependent variables, with which we are able to measure variation between cells, whereas others can be seen as independent variables, with which we are able to explain the observed variation. In dissociated single-cell profiling data, cellular phenotypes can be described based on dependent variables, such as overall gene expression, protein expression or chromatin accessibility. These dependent variables are a function of independent variables such as sample covariates, assay technology or sequencing batch. Spatial models of cellular variation extend this model by incorporating additional dependent variables, such as subcellular spatial distribution of molecules and cell morphology, and independent variables, such as spatial context of the cell, opening directions to new formulations of causal models and representation learning. Notably, the complementary view, with cells as independent variables ‘generating’ the tissue is just as biologically compelling, but it is not our focus here.

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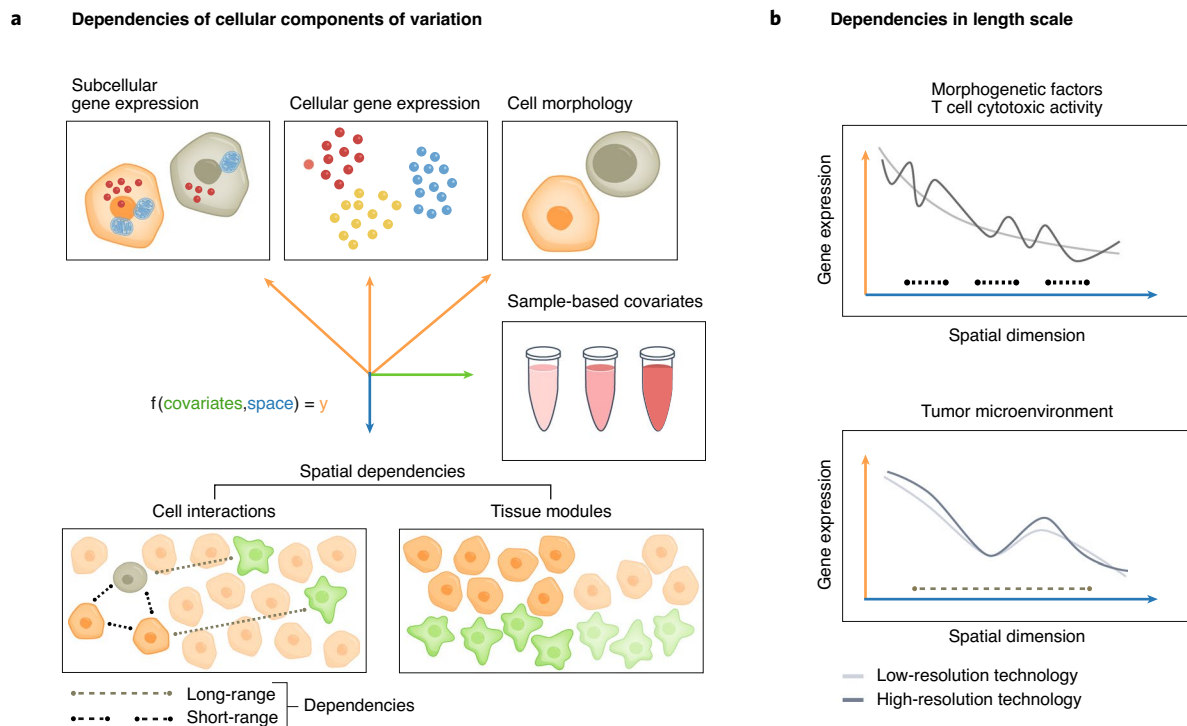


Fig. 1 | Components of variation in spatial profiling assays. **a**, Molecular profiles from spatial technologies entail dissociation-based measurements such as gene expression and sample-based covariates, as well as additional measurements such as subcellular variation, cell morphology, multimodal measurements in spatial context and spatial coordinates. In the figure, we highlight how such systems entail both long- and short-range effects that should be accounted for in modeling molecular variation in spatial dimensions. **b**, The study of several biological phenomena such as morphogenetic gradients in development, T cell cytotoxic activity and the tumor microenvironment can greatly benefit from the use of spatial technologies. In this figure, we also discriminate between observed variables (in solid lines) and latent variables (biological processes, dashed lines) that have to be unveiled from the data. Finally, the resolution of spatial technologies is also different and should be taken into account in the context of interest.

We distinguish two main influences on cell states in the spatial context ('dependency classes'): direct cell–cell communication and tissue context (Fig. 1). These dependency classes appear as high- and low-frequency patterns in a variance decomposition across a tissue, where the frequency of the pattern mirrors the length scale of the biochemical phenomenon underlying the statistical dependency. High-frequency patterns can occur multiple times within the same tissue but are localized to the direct neighborhood of a cell, and often correspond to cell–cell communication. Low-frequency patterns can represent global processes, such as developmental gradients in tissue ontogeny or oxygen availability gradients and other physiological characteristics. In a pathological setting, such as a tumor, a low-frequency component could be associated with signals related to metabolic gradients, while a high-frequency component could be associated with T cell cytotoxic activity (Fig. 1b). As is often the case in biology, we expect nesting as well within scale. For example, within a cell, we expect both local patterns (molecular complexes and liquid–liquid phase separated 'membraneless' organization), intermediate ones (organelles) and global ones (gradients in epithelia). Below, we stratify spatial effects and models by cell biology and tissue biology phenomena.

Cell biology reflected in spatial variation at cellular and subcellular resolution. In situ cell biology has largely been investigated with microscopy-based techniques, usually by visualizing a limited number of molecular proteins or RNAs¹³, whereas single-cell profiling has characterized the molecular features of cell states and types and their underlying circuitry at unprecedented scale and molecular detail in dissociated samples^{14–16}. Spatial molecular profiling

offers the possibility of unifying these two worlds of cell biology and molecular biology, thus advancing our understanding in two core areas: cell state variation and extrinsic effectors of cell states.

Molecular information collected with subcellular resolution characterizes cell states both by molecular abundance and the molecular distribution across the cell: subcellular features can also extend the cell state space and can disentangle cell states that cannot be distinguished based on cell-level molecular features in unsupervised analyses (Fig. 1a). Among the phenomena that can only be disentangled by subcellular-resolved molecular features are the effect of epidermal growth factor signaling on the cell cycle¹⁷ and organization of the transcriptome in the nucleus¹⁸ and in the central nervous system^{3,19}. In practice, if pixel-level molecular profiles are available, cellular phenotypes can be defined based on both organelle properties (for example, nucleus size) and the distribution characteristics of molecules across the cell, including their covariance structure (to define complexes, membraneless compartments and so on). Subcellular features of cells can yield hypotheses on their functional states, where the distribution of proteins and messenger RNA in a cell hints to organelle or pathway activity.

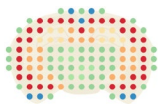
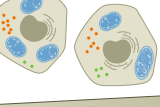
Spatial assays are also used to profile the effects of cell–cell communication occurring at short length scales (μm) in cellular neighborhoods^{20,21} via physical contact, compound exchange via tight junctions and synapses, and paracrine or autocrine signaling. Although single-cell profiling methods have been adapted to capture some direct cell–cell interactions^{22–24}, many communication modes in tissues can currently only be directly observed by spatial protocols. These interactions between neighboring cells have the potential to explain the variation observed within cell types in

Box 1 | Descriptions

Spatial molecular profiling technologies differ in terms of resolution and multiplexing. Therefore, the capability of the technology should be accounted for by modeling approaches. Here, we show how techniques can be grouped together based on the molecular entity they are able to capture, as well as the resolution (and therefore length scale) that they are able to profile.

- Transcriptome: spot-based spatially barcoded microarrays that capture transcriptome-wide gene-expression profiles: low resolution (spatial transcriptomics, Visium, Slide-seqV1) and high resolution (HDST, Slide-seqV2). In situ transcriptome profiles with barcoded oligos: high-throughput (MERFISH, seqFISH, NanoString), low throughput (SCRINSHOT (CARTANA), Barista-seq, osmFISH), sequencing by synthesis in situ (STARMAP) and others.
- Proteins: IMC (matrix-assisted laser desorption/ionization–time of flight, multiplexed ion beam imaging–time of flight) and multiplexed IHC (4i, cyCIF, CODEX, Immuno-SABER).
- Metabolites: imaging mass spectrometry.

Several reviews have covered experimental techniques for spatial molecular profiling^{217–229}.

Spots	Single-cell	Subcellular
		
Resolution		
Transcriptome		
• Spot-based (ST, Visium, Slide-seq) • Laser-capture microdissection	• Spot-based (HDST, Slide-seq V2, DBIT-seq, Stereo-seq, ...) • FISH-based and padlock-based methods	• FISH-based and padlock-based methods (seq-FISH-plus, MERFISH, STARmap, ...)
Proteins / metabolites		
• Imaging mass spectrometry	• Multiplexed IHC (CODEX, cycif) • IMC/spectrometry	• Multiplexed IHC (4i, ...)

dissociation-based protocols. For example, activated immune cell states can be understood based on the cells' neighborhoods, and correspond to variation observed in single-cell RNA-sequencing (scRNA-seq) data^{25,26}. It is important to carefully consider the experimental techniques used to probe cell–cell communication, as this inference depends both on the molecular features measured and whether true single-cell resolution is achieved (Fig. 1b,c and Box 1).

Inference of cell–cell communication (a functional feature) from molecular profiling data has been addressed by statistical association methods that typically relate cell types to each other via previous knowledge annotation^{27–30}, or by covariation patterns across samples^{26,31}. A spatial neighborhood provides strong evidence of co-occurring molecular and structural phenotypes, and should therefore be encoded as a strong prior in any model of tissue biology^{29,30,32,33}. Information on cell–cell communication also allows the development of causal inference frameworks, where directional models for cell state dependencies resulting from cell–cell communication in spatial neighborhoods may be used to yield causal and mechanistic hypotheses. Causal inference frameworks can, for example, explain the effect that a T cell has on a tumor cell^{34,35}, or how the metabolic states of the tumor microenvironment reshape

protein expression of immune cells³⁶. A causally interpretable model of these extrinsic effects in the microenvironment requires dependencies between cells and violates the independence assumption commonly used in cell-centric (intrinsic) models of gene expression in many analyses of scRNA-seq^{37–39}. An encoding of spatial proximity of cells as a spatial graph^{40,41} is one promising prior for such cell–cell dependency models.

Finally, recent advances in pooled perturbation screens with high content readouts, either optically^{42,43} or by single-cell profiling^{16,44}, open the way to inform and test such causal models, by observing both cell biology and molecular features under a large number of perturbations in cell cultures or in a tissue. An orthogonal approach to large-scale perturbation modeling with spatial information would be to selectively perturb cells in situ in spatially constrained regions of the system (for example, by light activation in animal models), to observe the perturbation effect on cellular communication and tissue architecture⁴⁵.

Tissue biology reflected in spatial variation on super-cellular length scales. Deciphering the emergent properties of tissue, the modular, multicellular units of tissue function at different scales, and their dysregulation in disease will require spatial technologies with larger fields of view, together with computational methods that can identify functional units and structures.

These 'tissue modules' are constituted by recurrent cellular communities that exert specific functions in tissue are spatially organized to execute this function and may occur at different sites in varying compositions. Such structures emerge at different length scales, and therefore can be very different in terms of size and cell composition. For instance, both germinal centers in lymph nodes and glomeruli in the kidney can be viewed as tissue modules, although they are very different structures in terms of size and function.

Tissue modules are defined by phenotypes that characterize areas of a tissue, as opposed to cell- or sample-centric phenotypes that are often the focus of models of dissociation-based data. Clearly, models for identifying tissue-level modules require molecular profiles as well as spatial resolution^{46,47}. However, spatial length scales are difficult to account for in such models. First, tissue modules are increasingly difficult to pinpoint as the length scale and the number of putative structures increase, because the relevant biological processes are likely to be convoluted with each other or with other co-occurring signatures. Second, the resolution of experimental methods often decreases as the field of view increases in size (for example, imaging time becomes limiting when imaging large volumes at high resolution) so that the potential for discovery of large and small structures is often anticorrelated. Third, the relation between length scales and functional structures or cellular communities will be different across tissues: the stereotypical layered organization of brains is different from the globular structure of the pancreas or the tree-like structure of lungs. Cross-length-scale models that work in one organ might not be useful for another organ.

Metabolic and morphogenetic gradients^{48–50} convey information about and arise from developmental processes as well as environmental differences, such as oxygen or nutrient availability, hormone concentration, metabolite concentration or physical stress⁴⁹. They can be assayed with most of the current spatial techniques, as they do not strictly require single-cell resolution. Additionally, biophysical forces, such as the stiffness of the extracellular matrix and its interaction with cell membrane, or cell densities in specific tissues⁵¹, could be inferred with spatial molecular techniques in combination with morphological features. Biophysical tissue features are relevant in many biological systems, from development to cancer biology^{51,52}, and they occur at multiple length scales. Coupling those phenotypic observations with molecular changes assayed by spatial omics has the potential to explain molecular determinants of mechanobiology.

Tissue-wide phenomena can be modeled via spatial variance decompositions and other unsupervised techniques^{53–58}. Clustering approaches that account for spatial proximity⁴⁶ or morphological similarity⁵⁹ could also be used to discover tissue models that go beyond molecular similarity. Unsupervised representation learning has been tremendously successful and widely used for scRNA-seq analyses^{37,60}. Advanced representation learning approaches can improve unsupervised models and offer promising tools to relate spatial patterns directly to tissue properties. These include established approaches in image-based deep learning that can deal with the absence of sufficiently informative labels, such as unsupervised, self-supervised and multi-task learning⁶¹. These techniques will be increasingly used for jointly modeling extrinsic features of cells from images, such as neighborhood density and morphology, as well as intrinsic features, such as expression profiles, thus learning a joint latent representation that encodes information from both modalities⁶². Biology-centered representation learning can be directly performed on image-structured data, which is often the raw version of spatial molecular profiling assays. However, this neglects the strong previous knowledge that cells are discrete functional units in tissues. To leverage this information, we can represent tissue as a spatial graph of cells (nodes) and their proximity (edges). Representation learning on images and graphs relies on very different model classes: graph-based models will often have interpretability advantages as they are defined on an interpretable input space of cells rather than pixels, analogous to capturing ‘social networks’ of cells with cell–cell communication and larger cliques. However, information aggregation across graphs is not straightforward^{63,64}, posing limitations for representing large-scale effects and could miss some of the continuous nature of the tissue, as reflected in images.

Whether such tissue modules are identified by unsupervised approaches or expert annotation, methods are necessary to harmonize them and coherently join them across samples and individuals to validate pattern discoveries. This validation requires a common coordinate framework for molecular variation across organs and individuals in which tissue modules can be registered across samples⁶⁵.

Tissue-level models of cell-fate decision-making. A key question addressed with single-cell genomic is inference of cell-fate cell lineage maps: arranging cells from one or more snapshots into continuous transitions across the molecular state space that explain cell-fate decision-making⁶⁶, as well as defining which cells are related by shared lineage through cell division events. Current spatial profiling techniques face the same issue of observing only a snapshot of biological processes, but they capture substantial auxiliary information that affects cell-fate decisions, cell lineage and cell differentiation; given the limited movement of cells in tissue in many contexts, physical proximity often reflects relatedness (for example, between stem cells in an intestinal crypt and their progeny in the villus), as well as the dependence of cells’ differentiation on external signals provided by neighboring cells and morphogenetic gradients (for example, between stromal cells and stem cells in the crypt).

Analysis of single-cell profiles has enhanced our understanding of cell-fate decisions and cell differentiation. Methods developed for dissociation-based data can be directly applied to spatial data, thus providing trajectory visualization in a spatial context^{2,67,68}. Moreover, spatial datasets allow cell-fate-inference methods to interpolate in not only the molecular cell state space, but also the spatial coordinates, with the potential to resolve subtle differences in cellular fates that were previously hidden (Fig. 2). From a causal modeling perspective, branching events might be explained by the spatial context; in particular, it may be possible to infer the paracrine signals affecting developmental processes^{69,70}, readily moving from a cell-intrinsic view of differentiation to a more comprehensive tissue view. Developmental processes where morphogenic factor localization

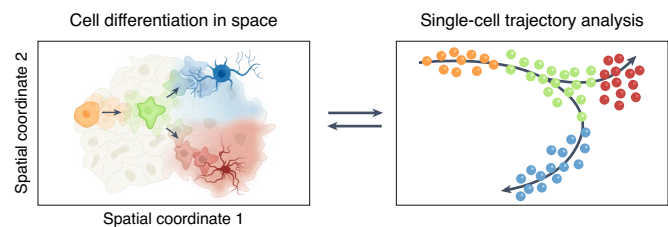


Fig. 2 | Trajectory inference in new dimensions. Spatial data allow trajectories inferred from the transcriptome to be placed in spatial context. It also opens up analytical approaches to infer trajectories with additional spatial information.

is tightly regulated on large length scales, such as in *Drosophila* embryos⁷¹, could be modeled by additionally accounting for cell–cell interactions and other spatial effects on short length scales. Overall, the spatial context of cell-fate decision-making could help to disentangle intrinsic and extrinsic effects of cell differentiation and developmental processes^{72–75}.

To determine cell lineages, efforts to integrate spatial molecular information with lineage-tracing techniques in engineered models^{76–78} and native human tissue⁷⁹ can also spur the development of new modeling approaches to disentangle spatial contribution to lineage formation. For instance, a recent study⁸⁰ has used lineage barcoding experiments in cerebral organoids to show that cellular clones have a strong regionalization during development and accumulate in specific brain regions. Another study⁸¹ has developed a lineage barcoding system suitable for spatial imaging techniques, such as multiplexed FISH, and applied it to the *Drosophila* brain, observing that gene-expression similarity between different clones was not related to spatial proximity, but that cells from the same lineage (that is, members of the same clone) had higher expression similarity when proximal in space than cells from the same lineage that were farther apart. With such data, lineage tracing can be used not only to validate proposed lineage relationships but also to constrain inference models and infer intrinsic and extrinsic factors of cell identity.

A key innovation in the field of trajectory inference from single-cell profiles has been the discovery of molecular states that relate to velocities in the state space (RNA velocity^{67,82}). Similarly, RNA velocity could be directly measured in situ. For example, MERFISH could determine RNA velocity by distinguishing intranuclear from cytoplasmic concentrations for specific transcripts². These state-space velocities could again be coupled to biological axes of variation. In a spatio-molecular lineage, one could extend molecular state-space velocities with spatial velocities of migrating cells. Taken together, these examples highlight the opportunities to directly translate, extend or reformulate concepts developed for single-cell genomics to spatial molecular profiling.

Spatial signatures of disease

Most disease processes involve the complex ecosystem of the entire tissue, and histology is accordingly a cornerstone of pathological characterization and definition of disease. Advances in single-cell profiling have highlighted the aberrations in cell states, compositions and interactions that are associated with disease onset, progression or response to therapy^{83,84} and have begun to delineate the coordinated changes across them^{34,35}. Spatial information can complement these signatures, explain the changes observed in dissociation-based protocols, disentangle additional alterations of tissue organization and relate them to the rich legacy of histopathology, which underlies both biomedical research and patient care. In addition, disease and treatment labels provide strong indicators of emergent properties of a tissue, such as tissue modules, and help in their discovery.

Spatial dimensions of disease state entail both tissue-scale deranged cellular communities and short-range perturbed cell–cell communications, therefore requiring models tailored to capturing effects at the relevant length scales. Several recent studies follow this approach in conditions as diverse as cancer^{34,35,85–89}, myocardial infarction⁹⁰ and neurodegenerative diseases⁹¹. For example, a recent study reported immune cell distributional signatures of T cells in colorectal cancer that do not manifest on a compositional level, at least for broad cell categories³⁵. Specifically, T cell aggregation correlated with core tumor phenotypes, whereas T cell number did not. Another study applied spatial transcriptomics to biopsies from prostate cancer tumors and identified cancer foci as multicellular subtissue compartments, often not identified by typical histopathological analysis⁸⁷. An extensive multicenter study of breast cancer tissue showed how cellular communities identified by IMC explain inter- and intrasample variation, as well as define cancer subtypes with strong associations to patient survival⁹². These communities, newly discovered tissue modules, have characteristic cell-type enrichments and putative cell–cell interactions. An integrative analysis of scRNA-seq and spatial transcriptomics of the same pancreatic cancer tumors, showed colocalization of stress-response cancer cells and inflammatory fibroblasts⁸⁶ forming a cancer-specific cellular neighborhood. Such studies are also relevant in nononcological pathologies. For example, spatial transcriptomics and FISH-based measurements of brain sections from a mouse model of Alzheimer's disease showed a local, multicellular interplay in cellular response to amyloid plaques⁹¹. These examples show how spatial information can both explain new disease features, and highlight the potential causal mechanism that connects a deranged cellular state to diseased tissue.

Analytical challenges and current approaches

So far, we have described how spatial components of variation can be mapped to cell and tissue biology questions, and highlighted some of the modeling approaches to disentangle spatial components of molecular variation. Next, we describe methods and analysis strategies that have been already proposed, and the practical problems of working with spatial molecular data, in terms of processing, representation, integration and analysis. We highlight available solutions and the open challenges that still need to be addressed.

Image processing, segmentation, coordinate registration and data structure. In contrast to dissociation-based protocols, most spatial methods require a computational mapping of the measured molecules to their cellular origin, either by segmentation or deconvolution^{93–104}. It is likely that segmentation-based approaches will need to be fine-tuned for the data at hand, due to the high diversity in spatial technologies (both in terms of resolution, multiplexing and modality) as well as variation across tissues. Semantic segmentation methods that assign each pixel of an image to a label of interest, such as blood vessels, stroma or connective tissue, offer an orthogonal approach to annotate tissue regions by morphology features. Their representation could be used for downstream tasks, such as integration with molecular data¹⁰⁵. Benchmarks and evaluations of segmentation methods, as well as integrating them in a processing pipeline^{106–108} could help decision-making on nuclear versus cytoplasmic segmentation, different segmentation algorithms^{109,110} and experimental design consideration on segmentation-auxiliary membrane stains⁷⁴. For imaging-based spatial technologies, alternative segmentation-free approaches have emerged. Such methods are often based on maximum likelihood estimation and graph representation learning, and aim to find pseudo-cells based on spatial aggregates of molecular probes^{111–114}. They present a viable alternative when experimental settings hinder robust segmentation-based approaches.

In addition to the molecular profiles associated with segmented cells, images convey invaluable statistics on cell area, morphology and population density that can help uncover principles of tissue

organization and higher-order structure across tissue samples. Image analysis techniques are also suitable for registering multiple tissue samples to a common coordinate framework⁶⁵. For example, methods that leverage convolutional neural networks for feature extraction and sample mapping have already been proposed in the context of spatial molecular data^{115–118}. Furthermore, to build 3D molecular maps of complex tissues and organs, techniques such as image registration and stitching will be essential^{115–117,119–121}.

A key computational question concerns data representation: spatial dimensions can be maintained as vector coordinates or transformed in a graph, while image information, such as extracted features and segmentation masking, should be easily accessible and directly linked to observations. Most of the existing software for (spatial) single-cell analysis should accommodate such data types associated with segmented cells in its existing framework^{59,108,122–128}. Efficient and flexible integration of image-structured data is often still difficult in cell-centric workflows as both data handling infrastructure and algorithmic approaches are missing (Fig. 3a and Table 1).

Integrating modalities. Some spatial methods capture rich molecular information at the cost of resolution, but computational analysis can be used for deconvolution and mapping by integrating the spatial data with microscopy data or single-cell profiles that have been determined using dissociation-based methods. For example, spot-based protocols such as HDST¹², Slide-seq^{8,9} and spatial transcriptomics/Visium⁴ capture RNA or proteins from areas spanning more (or less) than one cell, without consideration of cell boundaries. When spots are smaller than average cells, computational analysis can reveal cell identities by clustering, as has been demonstrated in HDST¹² and is possible through grouping of spots by segmentation masks defined in bright-field microscopy (FISH-based protocols and IMC^{3,6,129}). If spots are larger than one cell (for example, Visium), deconvolution methods can relate transcriptomes measured in a spot to the cell profiles present in tissue (Fig. 3b and Table 1)^{130–139}. In contrast to deconvolution approaches, that explicitly model the spatial observation as an aggregate of cells, label projection methods have been developed to map the cellular states identified in scRNA-seq experiments to lower resolution spatial data^{86,140,141}. These methods transfer cell-type labels from scRNA-seq to spot transcriptomics by manifold alignment techniques¹⁴². New methods should include the additional information provided by spatial omics data, such as morphology features derived from imaging data and spatial neighbor graphs computed from spatial coordinates. Such additional information could include smoothing constraints on cell-type proportions in space or morphological similarity derived from imaging data of cellular niches. The histology stain collected by spatial transcriptomics methods is invaluable for such mappings as it can be used to derive nuclei densities under each spot array, as well as other constraints in terms of morphology-based cell variability. For example, a recent method accounts for nuclei densities derived from tissue and adopts a probabilistic framework to assign cell-type proportions^{116,143}.

A second challenge lies in the feature space mapping: probe-based protocols such as FISH and IMC yield many fewer features and, in the case of protein measurements, also a different modality from scRNA-seq, and several methods have been proposed for mapping unobserved gene-expression profiles in sparse spatial data by learning a correlation-based representation from single-cell references^{74,140,141,144–146}. Beyond feature imputation, multiple tools are focused on mapping single-cell transcriptomes to spatial coordinate systems, using the spatial data as a reference map^{71,147–149}.

Recently, new methods emerged that reconstruct the spatial arrangement of cells mainly based on scRNA-seq data with little or no spatially resolved experimental data^{29,150}. They rely only on a few markers to perform the mapping, based on the assumption that similarity in gene-expression space correlates with spatial distances, and

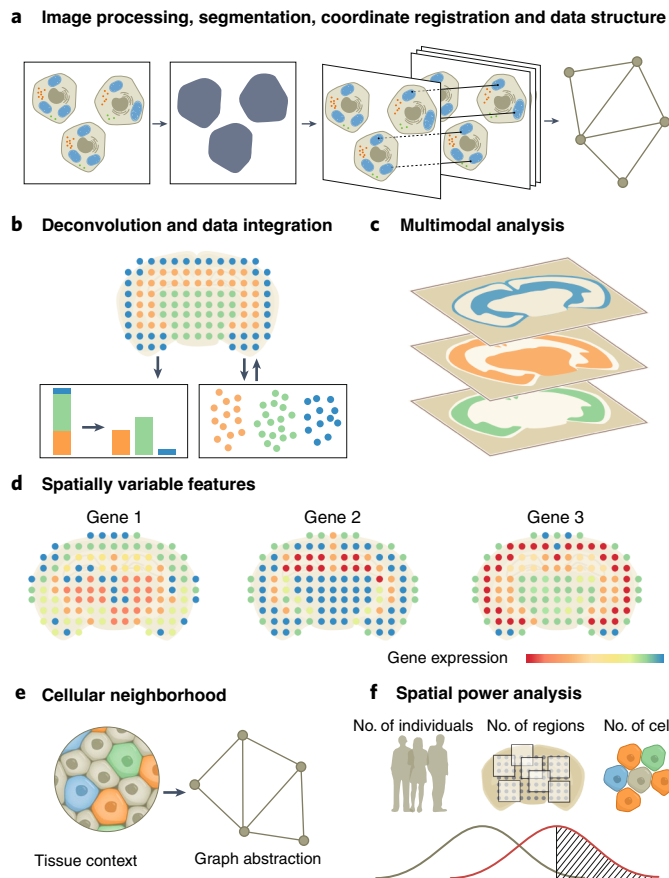


Fig. 3 | Analytical challenges and opportunities of spatial molecular data.

a, Image processing, segmentation, registration and data structure: spatial molecular data are diverse and require tailored solutions for processing and data infrastructure. **b**, Deconvolution and data integration: spot-based technologies require deconvolution analysis to approximate proportions of cell composition within spots. Integration methods can also be used to map known cell phenotypes to spatial data. **c**, Multimodality: integration of shapes and morphometric features, as well as additional molecular profiles improve cell state identification and tissue phenotyping. **d**, Spatially variable genes: regression frameworks that aim to find spatially variable molecular features are key in understanding cell states and tissue organization. **e**, Cellular neighborhoods: tissue coordinates are key in understanding spatial communities and cellular communication, and graph abstraction is a suitable representation of the data. **f**, Spatial power analysis: understanding tissue-level effects across samples and individuals requires power analysis that accounts for spatial distribution. Table 1 contains an overview of tasks and methods for spatial single-cell analysis.

that such a trend is monotonic in a defined neighborhood. While reference-free methods have been applied for biological systems where there is a strong stereotypical organization, for example due to global gradients (for example, *Drosophila* embryo), it is unclear how they perform in tissues that display disordered spatial heterogeneity, salt-and-pepper patterns and perturbation states, such as human organs and diseases. Learning such patterns from some spatial data, including the nature of cell–cell interactions, may help us find ways to introduce additional constraints for reference-free mapping, for example through receptor–ligand interactions. Mapping cell states to spatial coordinates is a challenging problem that we expect to remain an active area of research for other spatial molecular data.

Finally, there is a rich set of opportunities around integration of molecular and cellular information with the nonmolecular features

Table 1 | Challenges and current solutions for spatial single-cell analysis*

Challenge	References
Image processing, segmentation, registration and data structures	
Cell entity identification via segmentation	94–100,109
Cell entity identification via segmentation-free methods	111–113,179–182
Image processing	101–103,107,119,183–187
Image registration and alignment	115,116,119,126,188
Toolkit for spatial molecular data analysis	59,107,108,122,125–128,167,189–193
Deconvolution and multimodal integration	
Cellular-interaction inference	27,28,32,33,41,53,176,194
Data integration and deconvolution	130–134,136–139,141,143,195–199,200
Gene-expression mapping and imputation	104,144–146,154,201,202
Spatial mapping of gene-expression profiles	29,147,148,150,203
Multimodal integration	54,62,152,153,155,159,204,205
Spatially variable features, spatial communities and spatial power analysis	
Spatially variable genes	114,163–165,168–170,172,174,206–209
Spatial decomposition and clustering	46,47,54,56–59,114,162,171–173,182,210–213
Spatial power analysis	165,170,214–216

Tasks in the data analysis pipeline can be broadly divided into preprocessing, integration and spatially variable features identification. For each step in the analysis, we outline tasks and proposed solutions present in the literature.

of the image, to better define both cells and their tissue relationships (Fig. 3c and Table 1). Image analysis techniques and deep learning approaches can gather additional information from imaging data, including explicit or latent representation of cell area, morphology and population density¹⁵¹. These can help improve performances on different tasks such as cancer cell classification^{152,153} and mapping molecular counts at higher resolution¹⁵⁴. Multimodal representation learning that captures both imaging and molecular features can enhance our definition of cells (and neighborhoods), and allow training models that generate one modality from the other. On paired data, such models could be learned jointly yet should encode sources of variation separately across modalities, enhancing interpretability and biological discovery. When models are learned from uncoupled, separate data modalities⁶² (for example, a collection of microscopy images and of scRNA-seq profiles), joint measurements provide validation data with which to test predictions¹⁵⁵. Disjoint data sources are very likely to outmatch joint measurements in terms of datasets sizes (for example, H&E tissue slides for imaging and single-cell atlases for molecular data). For these very large datasets of different data types measured on different samples, (self-) supervised approaches such as contrastive learning could be used for pretraining and transfer learning^{156,157}. Learned representations could then be integrated ex post with interpretable linear methods^{158,159}. Such models could also be integrated with high-resolution imaging efforts, such as recently proposed high-resolution cellular maps^{160,161}. We forecast transfer learning efforts to be broadly useful for enriching legacy data (measured only in one modality) and facilitating future studies.

Spatially variable features and cellular neighborhoods. Although scRNA-seq has taught us how gene expression differs between individual cells, cell types and between different experimental conditions, spatial methods add another dimension to the analysis of gene-expression regulation. Several methods have been proposed to identify spatially variable genes in the tissue of interest (Fig. 3d and Table 1)^{162–167}. These methods vary greatly in terms of assumptions and model classes used. They can be broadly categorized into Gaussian process-based frameworks^{164,165,168–170}, methods that use hidden Markov random fields^{171–173} and graph-based methods^{114,174}. The power and sensitivity of these methods to detect spatial variability depends on the experimental technology and sample characteristics and thus has to be chosen carefully for the data at hand. Currently, the metrics that quantify spatial variability are still very heterogeneous because of the diversity in underlying algorithms, and this problem will need to be stratified in the future. Furthermore, established approaches such as differential expression analysis across conditions should be expanded to account for the variation of the spatial dependency over samples, as well as model latent spatial variance components of the data⁵³. We expect that techniques borrowed from the field of spatial statistics will also be useful to model spatial components of molecular variation, in particular accounting for different views of tissue organization: from direct cell–cell interaction to higher scale interactions such as paracrine communication to metabolic gradients.

Spatial techniques will also be useful to study cellular networks and cell–cell communications (Fig. 3e). Few methods have been proposed that aim to estimate cell-type neighborhood enrichments in spatial context^{59,106,175}, and inferring cell interaction and signaling genes from spatial graph structure^{33,41,58,59,106,172,176}. The modeled phenomena differ drastically between these methods, and will both need to be stratified in the future.

Spatial power analysis. Power analysis enables researchers to determine the sample sizes required to observe an effect of interest. In the context of spatial molecular data, such effects are subject not only to sample variation but also to its spatial coordinates in the samples. This is of particular importance in spatial systems since realistically attainable sample sizes (image size or image number) depend strongly on the chosen experimental protocol. For clarity, we propose to frame the problem in two subchallenges: gene-centric and cell-centric power analysis (Fig. 3f).

‘Gene-centric’ power analysis entails the power calculation to find significant spatially variable genes in tissue. In this context, each gene is considered separately and a power calculation can be performed based on an estimated effect size of gene variability in nonrandom regions of the tissue samples. A recent comprehensive power analysis, using synthetic datasets¹⁶⁵, evaluated several methods that rely on different test statistics^{162–165}. The superposition of multiple confounding spatial expression trends, such as high- and low-frequency patterns (Fig. 1b), further complicates trend discovery. Notably, asking an algorithm to identify any dependence of expression on spatial coordinates is often too general a question, whereas a user might be interested in finding patterns of a certain shape, such as radial symmetries in tumors. Such pattern-based trend discovery has previously been attempted in spatial variable gene detection, but may be limited by the hypothesis that can be framed as patterns^{164,165}.

‘Cell-centric’ power analysis concerns features of tissue organization, such as cellular neighborhoods, cellular communities and tissue composition. For instance, the optimal sample size (total number of cells) to observe a particular enrichment of cellular neighborhoods depends on the frequency of the participating cell types and the required significance threshold. Moreover, large-scale tissue features often correspond directly to images so that the sample size is the number of images obtained, posing tradeoffs between

number and size of images. This problem has been investigated in the context of dissociation-based assays^{177,178} but is yet unexplored for spatial molecular data.

Conclusions

Spatial profiling carries the promise of connecting variation in high-dimensional single-cell omics datasets with interpretable biological phenomena in tissue. These spatial dependencies exist on drastically different length scales and correspond to cell biological effects with different implications at tissue level. Here, we considered effects that can be modeled across these length scales and coupled the existing spatial profiling methods that provide pertinent measurements. We formulated a roadmap for modeling dependencies in spatial data based on the length scale of underlying molecular phenomena. We also classified current processing and analysis approaches to spatial molecular data. As highlighted by different application settings, such as spatial signatures of disease, models for spatial variation can resolve cellular states and tissue phenotypes that were previously hidden. Accordingly, model-based experimental design of spatial profiling studies is not only limited by resolution but is also a compromise among resolution, section size and sample size.

Data analysis currently still presents many bottlenecks, as opportunities inherent in integrating data from spatial to nonspatial and between spatial assays of different resolutions and for different analytes are not fully realized. Major challenges remain in the analysis of image-structured data in terms of cell segmentation, image processing and the relation of tissue structures observed in the image to the of the measured molecular profiles. New ways of thinking about processing imaging data in light of modeling particular length scales may alleviate data processing bottlenecks as spatial models move from pipelines to end-to-end models. On the other hand, mathematical models for length scale specific and cross-length-scale spatial variation have only started to emerge. Ultimately, we will find solutions to the remaining analytical and experimental problems that will allow the creation of single-cell resolved and spatially aware tissue atlases.

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Competing interests

F.J.T. reports receiving consulting fees from Immunai and ownership interest in Dermagnostix GmbH and Cellarity. A.R. is a founder of and equity holder in Celsius Therapeutics, is an equity holder in Immunitas Therapeutics, and was a scientific advisory board member for ThermoFisher Scientific, Syros Pharmaceuticals and Neogene Therapeutics until 1 August, 2020. Since 1 August, 2020, A.R. has been an employee of Genentech. A.R. is a named inventor on several patents and patent applications filed by the Broad Institute in the area of single-cell and spatial genomics.

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